

Your turn — reproduce a document

IATDMCT Congress · Introduction to Markdown, Quarto and equations

Markus Hovd

2026-09-25

1 The task

Write a Quarto document that renders like **reproduce-target.pdf**. That file is the specification: open it, read it, and build a `.qmd` that produces the same thing.

You are not being asked to reproduce it word for word. What matters is that every *mechanism* in it works — the numbering, the cross-references, and the numbers and figure that are computed rather than typed.

The input file is `data/patient-concentrations.csv`, in this folder.

2 What the document must contain

- ☐ A **YAML header** with `title`, `author`, `date: today` and PDF output
- ☐ **Numbered sections** using `##` headings
- ☐ A paragraph containing **bold**, *italic* and an **inline equation**
- ☐ A **numbered display equation**, given a label, and referred to in the text with `@eq-...` so that “Equation 1” appears
- ☐ An **R code chunk** that reads the CSV and draws a figure
- ☐ That figure **cross-referenced** with `@fig-...` so that “Figure 1” appears
- ☐ A **table** with a caption and a label, referred to with `@tbl-...`

3 How to work

Render after every change — that feedback loop *is* the skill:

```
quarto render my-report.qmd
```

If the render fails, read the error message. It names a file and a line. Nine times out of ten the cause is a YAML indent, an unmatched `$`, or a missing blank line.

💡 Stuck?

`reproduce-solution.pdf` is the same document with every code chunk shown, and `reproduce-solution.qmd` is its source. Try first, then compare — you will learn more from one failed render than from reading the answer.

4 What each part is really testing

Part	What it demonstrates
<code>@eq-...</code>	That a numbered equation is <i>referenced</i> , never typed by hand
The R chunk	That the document runs your code when it renders
<code>@fig-...</code>	That a figure is numbered and linked automatically
<code>@tbl-...</code>	The same for a table, with a caption
The inline equation	That math works in the middle of a sentence, not just on its own line

If you can do all five, you can write a reproducible report. That is the whole point of the session.